THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2012: Fundamental Concepts of Mathematics

May 12, 2022 2:30 p.m. – 5:00 p.m.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL SIX questions

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

Remark: Begin the answer to each question on a new page (sub-parts of a question can be put in the same page).

1. (24 points) Prove or disprove each of the following statements:

   (a) The product of any pair of twin primes, that is, primes of the form $(p, p+2)$ is congruent to 8 (mod 9) for any $p \geq 5$.

   (b) For any $m, n \in \mathbb{N}$ with $m > n$, if $\sqrt{m} + \sqrt{n}$ or $\sqrt{m} - \sqrt{n}$ is rational, then both m and n are perfect squares.

   (c) There exist nonempty sets A, B, C such that $A \times B \subseteq B \times C$ and $A \not\subseteq C$.

   (d) For any sets A, B, C, if $A \cup C$ is numerically equivalent to $B \cup C$, and $A \cap C = B \cap C = \emptyset$, then A is numerically equivalent to B.

2 (12 points)

   (a) Prove that if there exist $p, q, r \in \mathbb{N}$ satisfying the equation

$$p(p-5) + q(q-5) = r(r+5), \qquad (*)$$

   then at least one of p, q, r is a multiple of 5.

   (b) Hence, or otherwise, find all primes p, q, r satisfying $(*)$.

3. (16 points) Consider the sequence $F_1, F_2, F_3, \ldots$ of Fibonacci numbers satisfying $F_1 = 1$, $F_2 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for any integer $n \geq 3$.

   (a) Find all $n \in \mathbb{N}$ such that $F_n = n^2$.
   (*Hint*: Use induction to show that $F_n > n^2$ from a certain value $n = n_0$ onward.)

   (b) Let $\alpha = \frac{1+\sqrt{5}}{2}$. Express α^n in terms of α, F_{n-1} and F_n for any integer $n \geq 2$, and use induction to prove that your formula is correct.
   (*Hint*: Note that α is a root of the quadratic equation $x^2 - x - 1 = 0$.)

   (c) Use the results from (a) and (b) to find all $n \in \mathbb{N}$ such that $\alpha^n - n^2\alpha$ is an integer.
   (*Hint*: You may use without proof the fact that α is irrational.)

4. (18 points) Let A be a nonempty set and $f : A \longrightarrow A$ be a function. For any $n \in \mathbb{N}$, denote by f^n the composition of n times of f, that is,

$$f^n = \overbrace{f \circ f \circ \cdots \circ f}^{n \text{ times}}.$$

Define $f^0 = id_A$, the identity function on A.

Let R be a relation on A defined by the following: for any $a, b \in A$,

$$aRb \text{ if } [(a = f^n(b)) \text{ or } (b = f^n(a))] \quad \text{for some } n \in \mathbb{N} \cup \{0\}.$$

(a) Prove that R is reflexive and symmetric.

(b) Give an example of a set A and a function $f : A \longrightarrow A$ such that R is not transitive.

(c) Prove that if f is injective, then R is transitive (and hence is an equivalence relation on A).

(d) Let $A = \mathbb{Z}$ and $f : \mathbb{Z} \longrightarrow \mathbb{Z}$ be given by $f(m) = m + 2022$. Find all the distinct equivalence classes and hence give a partition of $\mathbb{Z}$ with respect to R.

5. (16 points) Let A and B be nonempty sets and let $f : A \longrightarrow B$ be a function. Consider the following statements:

(1) For any $Y \subseteq B$, there exists $X \subseteq A$ such that $Y = f(X)$.

(2) For any $X \subseteq A$, there exists $Y \subseteq B$ such that $X = f^{-1}(Y)$.

Exactly one of the above statements is a necessary and sufficient condition for "f is injective" and the other is a necessary and sufficient condition for "f is surjective". Make your assertions and prove each of them.

6. (14 points) Consider the set A of proper subsets S of $\mathbb{N}$ such that both S and $\mathbb{N} \setminus S$ are infinite sets, that is,

$$A = \{S \subset \mathbb{N} : S \text{ and } \mathbb{N} \setminus S \text{ are infinite sets}\}.$$

(a) Give an example of an element of A.

(b) Prove that A is itself an infinite set.

(*Hint*: You may use the fact that a set is infinite if it contains an infinite subset.)

(c) <u>Prove or disprove</u> that A is countable.

******* **END OF PAPER** *******